

Mason Vrshek

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Skills

CAD: Siemens NX, SolidWorks (CSWA), Teamcenter PDM

Programming/Controls: Matlab, Python, Java, ROS 2, Git

Education

Purdue University – B.S Mechanical Engineering

GPA: 3.97/4.00

May 2029

Experience

Undergraduate Research Asst., Mechanisms and Robotic Systems Lab – Purdue University Dec 2025 – Present

- Developed a sensor hardware stack and housing integrating a 76° FOV CMOS camera, reflective polariscope, photoelastic elastomer, and controllable illumination into a 7 mm-diameter resin-printed vision-tactile platform; modeled components in Siemens NX and validated structural designs using Fusion 360 FEA.
- Integrated a ResNet-18 + GRU classifier with tactile video data to distinguish exposed and subdermal tissue features with 97.4% and 92.1% accuracy
- Integrated MVP-Tac with a UR5e robotic platform for vision-guided 3D reconstruction and automated tactile palpation

Manufacturing Engineering Intern – Altec Industries May 2026 – Aug 2026

- Designed and released FEA validated components in SolidWorks for utility vehicle assemblies, managing full revision lifecycle through Teamcenter PDM to ensure BOM accuracy and change traceability across teams.
- Built SQL-backed Power Automate workflows to surface previously buried data, spanning 2000+ records per week and saving approximately 8 hours per week across 4 departments
- Built a Visual Management Television program using Powerquery, powerBI, and Excel to surface production KPI's such as cell throughput, quality infractions, and labor utilization enabling floor supervisors to identify workflow bottlenecks and make data-driven decisions without reliance on manual reporting

Unmanned Aerial Systems researcher, Vertically Integrated Projects – Purdue University Aug 2025 – May 2026

- Developed ROS 2 flight-control software for autonomous aerial systems and performed motion capture testing using Qualisys track manager software
- Used OpenSCAD to design a custom PCB mount to centralize electric components and easily shift aircraft center of mass, validating the resulting configuration through flight-test data and XFLR5 α -sweep simulations.

Projects

Active Orthosis, ASME: Biomechanics

- Implemented FEA-driven design improvements to a robotic orthosis chassis capable of producing >150 N of output force.
- Designed and routed a custom KiCad PCB, integrating sensor inputs and driver signals into a centralized control system

SnowBot

- Modeled an autonomous snow-plowing robot, developing the chassis, drivetrain, plow assembly, and mechanical integration of the robotic system.
- Architected an ESP32-based embedded control system integrating a BNO085 IMU, IBT-2 high-current motor driver, NEO-V1.1 motors, and 12V battery system for autonomous sensing and drivetrain control.
- Presented the proposed autonomous snow-plowing system, including CAD design and engineering analysis, to a panel of industry judges at Purdue's First-Year Engineering Design Competition.